

Name _____

Score _____

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!** Attach this sheet to the front of your answers.

1. (8 pts) Use the method of *Reduction of Order* to find a second linearly independent solution y_2 of the differential equation $x^2 y'' + 2xy' - 6y = 0$ given a known solution of $y_1 = x^2$.
2. (8 pts) Find the general solution of the differential equation $y'' + 4y' - 2y = 0$
3. (8 pts) Find the general solution of the differential equation $y'' - 4y' + 29y = 0$
4. (14 pts) Solve the initial-value problem $y'' + 6y' + 9y = 0$, $y(0) = -3$, $y'(0) = 1$
5. (14 pts) Find the general solution of the differential equation $y'' - 2y' - 8y = 2 - x^2$
6. (14 pts) Find the general solution of the differential equation $y'' - y = -6e^x$
7. (8 pts) Find the general solution of the differential equation $x^2 y'' + 5xy' + 3y = 0$
8. (8 pts) Find the general solution of the differential equation $x^2 y'' - 3xy' + 13y = 0$
9. (14 pts) Use the method of *Variation of Parameters* to find a particular solution of $x^2 y'' - 2xy' + 2y = \ln x$
10. (10 pts) A mass weighing 6 lbs is attached to a 4 ft spring and stretches the spring so that its new length is 7 ft. This mass is replaced by another mass which weighs 8 lbs. The mass is released at a point $\frac{1}{2}$ ft below the equilibrium position with an initial downward velocity of 1 ft/s. Find the equation of motion.